

A net-zero Gozo by 2040?

Testing the Nationalist Party's 2026 pledge against open data and a measured forest

THE CLAIM UNDER REVIEW

“A clear and realistic plan so that by 2040 Gozo becomes a Net-Zero Island.”

Partit Nazzjonalista, programme Nifs Ġdid 2026, chapter Ghawdex (our translation).
News summaries: Gozo would become net zero “through afforestation”.

PLEDGE: NOT MEASURABLE

As of 5 Oct 2026: no baseline, boundary or pathway published

Version 1.1 · 5 October 2026

Status: draft for right of reply (Partit Nazzjonalista)

Prepared from public sources and open data. No site visits.

Repository: github.com/leandergrech/Mizien

MIŻIEN · AN INDEPENDENT FACT-CHECKING PROJECT

Contest a verdict, with evidence: github.com/leandergrech/Mizien/issues

TL;DR

Before the general election of 30 May 2026 [7], the **Nationalist Party** promised “a clear and realistic plan” for **Gozo to become a Net-Zero Island by 2040**, with afforestation among the measures. News reports summarised it as net zero “through afforestation”. We tested whether the pledge has a baseline, a boundary and a pathway that would let anyone check it, and whether afforestation could do the job.

- 1

A plan to make a plan. The programme promises a plan but publishes no emissions baseline, no boundary (imported electricity, ferries, tourism) and no pathway to 2040.
- 2

There is no official inventory, but there is a baseline to build on. A 2023 study for the EU’s Clean energy for EU islands secretariat put Gozo’s energy-related CO₂ at 118,000–154,000 t a year in 2016–2020 [6]. Our population-based estimate is about 157,000 t.
- 3

Afforestation alone could not do it. Offsetting those emissions with new forest would need 4.9–6.3 times Gozo’s land area at a measured semi-arid rate (1.5–8.5 times across our wider range). The PN’s own text relies on cutting emissions first; the “through afforestation” version is a news summary.
- 4

Pledge label: not measurable (as of 5 October 2026). A pledge is not a statement of fact, so it is not false; as worded it cannot be shown to be achieved or missed.

THIS CLAIM

NOT
MEASURABLE

NOT YET
DUE

ON
TRACK

OFF
TRACK

MET

MISSED

118–154 kt

Gozo energy CO₂ a year,
2016–20 (EU islands study)

~157 kt

Our population-based estimate
(no official figure)

4.9–6.3×

Gozo’s area needed as new
forest to offset that (measured
rate)

2040

Target year; no published
pathway

WHAT WOULD MAKE IT MEASURABLE

A Gozo emissions inventory with a stated boundary and a costed 2040 pathway showing the shares from cuts, afforestation and offsets.

WHAT WOULD COUNT AGAINST IT

A net-zero claim for Gozo that relies on unspecified offsets, or leaves out ferries or imported electricity without saying so.

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Limitations

1 The pledge and what we could verify

Our candidate record took the pledge from a MaltaToday summary [2]. We then found the PN’s own text [1], in Maltese. The translation is ours; the original wording is quoted so readers can check it. The text sets a 2040 date and lists afforestation as one measure among many, which the summary did not.

Source	What it says (our translation)	Our access	Status
Partit Nazzjonalista, programme <i>Nifs Ġdid</i> , chapter Ghawdex [1]	“Infasslu pjan ċar u realistiku biex sal-2040, Ghawdex isir Net-Zero Island...” <i>We will draw up a clear and realistic plan so that by 2040 Gozo becomes a Net-Zero Island, the first in the Mediterranean where every unit of emissions is reduced or offset, through solar panels, electric cars, taxis and vans, building efficiency, ecological restoration on land and at sea, afforestation, and water and soil conservation, among others.</i>	Read in full in a browser.	The pledge
MaltaToday, May 2026 [2]	Gozo “would become a net-zero island through afforestation”.	Read (Wayback copy).	Summary

FAIRNESS NOTE

Labour won the general election of 30 May 2026 [7], so this is an opposition proposal; Labour’s own environmental pledge from the same campaign is Claim Check 011, held to the same standard. Island decarbonisation is EU policy, and an energy baseline for Gozo already exists [6] that a plan could start from. This check is about whether the pledge can be measured. Until 5 October 2026 it was part of Claim Check 011.

2 Method

Question. Does the pledge have a baseline, boundary and counting method that would allow it to be checked, and is net zero through afforestation physically plausible on Gozo?

Gozo’s emissions. No official greenhouse-gas inventory exists for Gozo. We used two estimates: (a) population (41,253 on 1 January 2025 [4]) times Malta’s national emissions per person (2.5, 3.81 and 5.0 t CO₂e; 3.81 t is Malta’s 2024 figure in Claim Check 003 [8]); and (b) the energy-related CO₂ estimated for 2016–2020 in the 2023 Energy Baseline Scenario for Gozo [6] (electricity, transport on the island, ferries to and from it, heating and cooling).

Sequestration. A measured rate from a semi-arid Aleppo pine afforestation on rendzina over limestone (about 3.6 t CO₂/ha/yr over 35 years [3]) and an optimistic 10 t. Gozo’s area is 67 km² [5]. Code: `tools/cc-107-report/gozo_calc.py`; results in `data/cc-107/gozo_net_zero_arithmetic.csv`.

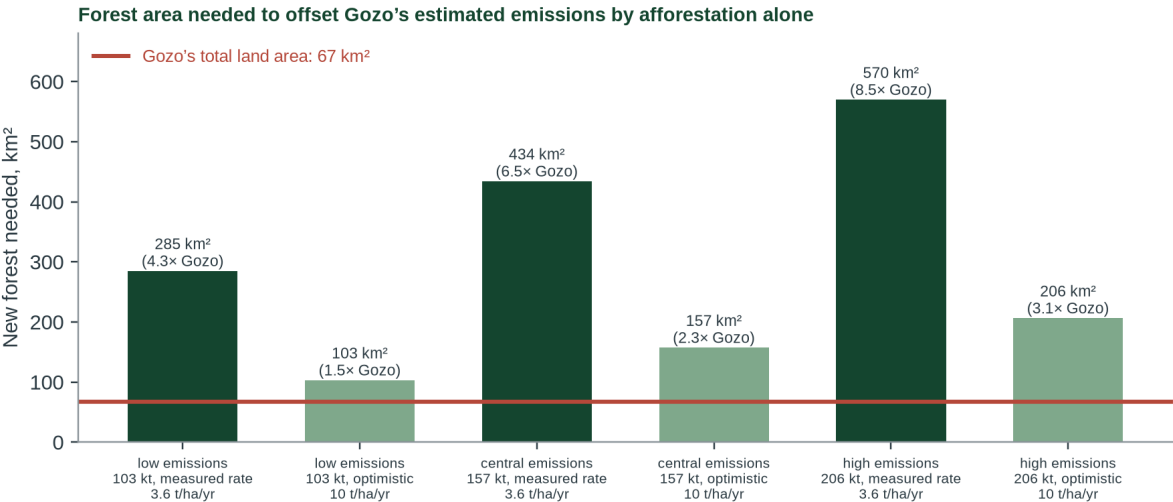
Grades. The afforestation inventory is B; official statistics and the EU islands study are C; our own arithmetic is an order-of-magnitude estimate and is labelled as such.

3 What “net zero” would have to mean

The pledge rests on a term that is not defined for Gozo. Which emissions count? The 2023 baseline counts electricity drawn from Malta’s grid, transport on the island and the ferries [6]; transport alone is 60–70% of Gozo’s energy use there. A boundary that left out the ferries or imported electricity would make the target easier without any change on the island. And what counts as an offset: forests on Gozo, restoration at sea, or credits bought elsewhere? The programme does not say.

4

The arithmetic



Emissions = Gozo population (41,253) × 2.5 / 3.81 / 5.0 t per person. Measured rate: Grünzweig et al. 2007 (Aleppo pine, 35 years). Source: data/cc-107/gozo_net_zero_arithmetic.csv.

Figure 1. New forest needed to offset Gozo's estimated annual emissions by afforestation alone, under three population-based emissions estimates and two sequestration rates, compared with Gozo's land area.

Emissions estimate	t CO ₂ a year	Forest needed, measured rate	Multiple of Gozo
EU islands baseline, lowest year (2020) [6]	118,333	327 km²	4.9×
EU islands baseline, highest year (2019) [6]	153,997	425 km²	6.3×
Population × 3.81 t (central) [4, 8]	157,174	434 km²	6.5×
Population × 2.5 t, optimistic 10 t/ha/yr	103,133	103 km²	1.5×
Population × 5.0 t, measured rate	206,265	570 km²	8.5×

Measured rate 3.62 t CO₂/ha/yr [3]; Gozo 67 km² [5]. Full results in data/cc-107/gozo_net_zero_arithmetic.csv.

THE ARITHMETIC

Gozo population 41,253 × 3.81 t per person = about 157,000 t CO₂e a year, close to the highest year in the 2023 energy baseline (154,000 t in 2019). A measured semi-arid pine afforestation stored about 3.6 t CO₂ per hectare per year [3]. Offsetting 118,000–154,000 t would take 33,000–43,000 ha of new forest: 4.9 to 6.3 times Gozo's 67 km² [5]. Planting 10% of Gozo would offset about 1.5% of the central estimate.

What this means for the pledge

- **“Net zero through afforestation”, as reported, is not possible on Gozo.** But that is the news summary [2], not the PN’s text.
- **The PN’s text relies on cutting emissions first** (solar, electric vehicles, efficient buildings), with restoration and afforestation as part of the mix. Whether that reaches net zero by 2040 depends on numbers the programme does not give: a baseline, a boundary, and the role of offsets bought from elsewhere.
- **Sequestration in young plantations is slow at first and can be reversed** by drought or fire; the measured rate [3] is a 35-year average on a site with deep groundwater, not a guaranteed yield on Gozo’s thin soils.
- **“The first island in the Mediterranean”** was not tested.

5 Testing the pledge

Sub-claim	Said by	What the evidence shows	Rating
A. A clear and realistic plan for a Net-Zero Gozo by 2040	PN [1]	No plan published; no official inventory or boundary. A 2023 energy baseline exists [6] but the programme does not use or cite it. Cannot be assessed as “realistic”.	NO BASELINE
B. Gozo net zero through afforestation (news summary)	MaltaToday [2]	Would need 4.9–6.3 times Gozo’s area as new forest at a measured rate. Not the PN’s own wording.	NOT POSSIBLE AS REPORTED
C. The first island in the Mediterranean where every unit of emissions is reduced or offset	PN [1]	Not tested.	NOT TESTED

6 Pledge label and requests for evidence

PLEDGE

NOT MEASURABLE

As of 5 October 2026. A plan promised without a baseline, boundary or pathway; afforestation alone would need 4.9–6.3 times Gozo’s area.

Why. The pledge promises a plan for an outcome whose starting point and boundary are not defined, and the widely reported version (afforestation) is physically impossible on Gozo’s land area. A pledge is not a statement of fact, so we do not call it false. We label it *Not measurable*: as worded, it cannot be shown to be achieved or missed. Pledges get one of six labels rather than a verdict (Appendix A).

What this label does not say. It does not say the goal is undesirable, or that anyone acted in bad faith. A checkable version would read: “Gozo’s emissions, on [boundary], were [X] t in [year]; by 2040 we will cut them to [Y] t and offset the rest by [means].”

Evidence we are asking for

- ☐ From the PN: a Gozo emissions inventory and its boundary (imported electricity, inter-island ferries, tourism).
- ☐ From the PN: the share of the 2040 target expected from cuts, from afforestation and from offsets elsewhere.
- ☐ From the PN: the area to be afforested on Gozo and the species and sequestration rate assumed.
- ☐ From the PN: whether the plan would start from the 2023 Energy Baseline Scenario for Gozo.

RIGHT OF REPLY

Before wider circulation this draft should be sent to the Partit Nazzjonalista with a fixed deadline (suggested 14 days). Responses will be appended and the label revisited.

7 Limitations

- The population-based estimate uses national per-person figures; Gozo’s real profile (less industry, more tourism and ferry traffic) may differ. The 2.5–5.0 t range is our assumption.

- The 2023 baseline is a study prepared for the EU islands secretariat, not official statistics. It covers energy use only, attributes grid electricity with conversion factors, and reflects its authors' views, not the Commission's [6].
- The sequestration rate comes from one well-studied site in Israel; no Maltese measurements were found.
- The programme translation is ours.

References

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- [1] Partit Nazzjonalista (2026). *Nifs Ġdid: Programm Elettorali*, chapter Ghawdex. <https://pn.org.mt/en/nifsgdid/ghawdex/>
- [2] MaltaToday (May 2026). PN approves its electoral manifesto. https://www.maltatoday.com.mt/news/election-2026/141891/pn_approves_its_electoral_manifesto
- [3] Grünzweig J.M., Gelfand I., Fried Y., Yakir D. (2007). Biogeochemical factors contributing to enhanced carbon storage following afforestation of a semi-arid shrubland. *Biogeosciences* 4(5):891–904. doi:10.5194/bg-4-891-2007. (Full text read; open access.) <https://doi.org/10.5194/bg-4-891-2007>
- [4] Eurostat. Population on 1 January by NUTS 3 region (demo_r_pjanaggr3), MT002 Gozo and Comino, 2025. Retrieved 2 Oct 2026. https://ec.europa.eu/eurostat/databrowser/view/demo_r_pjanaggr3/default/table
- [5] European Commission. Clean energy for EU islands: Malta (Gozo area 67 km²). <https://clean-energy-islands.ec.europa.eu/countries/malta>
- [6] Vaz L., Rodrigues de Almeida J. (2023). *Clean energy for EU islands: Energy Baseline Scenario for Gozo*. Clean energy for EU islands secretariat, 25 January 2023. Table 16. (Read in full.) https://clean-energy-islands.ec.europa.eu/system/files/2025-06/REPORT_TechnicalAssistance_Gozo_20230223.pdf
- [7] IFES ElectionGuide. Malta: Maltese House of Representatives 2026 General (held 30 May 2026; results source: Electoral Commission of Malta). Accessed 5 Oct 2026. <https://electionguide.org/elections/id/5161/>
- [8] Miżien. Analysis code and outputs: tools/cc-107-report/; data/cc-107/; Malta's 2024 emissions per person from data/cc-003/checks.csv.

Appendix A: series standards and verdict scale

Every investigation in this series follows the same rules so that verdicts are comparable and challengeable.

Verdict	Meaning
SUPPORTED	Evidence consistently backs the claim as stated.
LARGELY SUPPORTED	Backed by the evidence, with minor caveats that do not change the substance.
NOT SUBSTANTIATED	Stated more strongly than the evidence offered or available allows. May be true in some conditions but has not been shown.
MISLEADING	Omits material facts so that the overall impression is inaccurate, even if individual statements are defensible.
CONTRADICTED	The available evidence points against the claim.

Pledge labels

A pledge is a promise of future action, so it cannot be true or false when it is made. It gets one of these labels instead of a verdict, with the date of the evidence behind it.

Pledge label	Meaning
NOT MEASURABLE	The pledge as worded has no definition, baseline or date against which delivery could be checked.
NOT YET DUE	Measurable; the deadline or term has not passed and no progress data have been published yet.
ON TRACK	Published progress is at or ahead of a straight-line path (or the pledge's own milestones) to the target by the deadline.
OFF TRACK	Published progress is behind that path, and the target has not yet been shown to be met or missed.
MET	Documents or data show the target was reached, by the deadline if one was set.
MISSED	The deadline or term has passed, and documents or data that can be shown say the target was not reached.

Confidence

High: multiple independent lines of evidence agree. **Moderate:** evidence is relevant but incomplete or indirect. **Low:** evidence is thin, second-hand or conflicting.

Evidence grades

A experiment · B observational study with a control or gradient · C review, guidance or official statistics · D assertion or anecdote. Our own arithmetic is an order-of-magnitude estimate.

Standards

- **Quote the speaker, not the headline**, with outlet and date, and link the original where possible.
- **Separate the claim from the evidence offered for it**, and say who made which statement.
- **Grade the evidence** and show differing views side by side where the evidence is unsettled.
- **State what we could not access or verify**, and what would change the verdict either way.
- **Ask for the missing evidence** in specific, answerable requests.

- **Offer a right of reply** before wider circulation and append responses.
- **Correct errors openly** with a version number and change log.
- **Whistle-blow only on evidence:** a verdict of *Misleading* or *Contradicted* requires documents or data that can be shown, not inference.
- **Check every side.** Government, opposition, NGOs and developers are held to the same standard.

Appendix B: revision log

Version	Date	Change
1.0	5 Oct 2026	First issue as a separate check. The analysis first appeared as pledge 2 of Claim Check 011 (v1.0, 2 Oct 2026; v1.1, 5 Oct 2026). New in this issue: the 2023 Energy Baseline Scenario for Gozo [6], which gives an energy-emissions baseline and puts the forest needed at 4.9–6.3 times Gozo’s area at the measured rate (the 1.5–8.5 times range is kept). Verdict as in Claim Check 011: not substantiated, moderate confidence. Draft pending right of reply from the Partit Nazzjonalista.
1.1	5 Oct 2026	Maintainer decision (5 Oct 2026): pledges get one of six pledge labels instead of a verdict (methodology, Pledges). This pure pledge check is now labelled “Not measurable” as of 5 October 2026 in place of the verdict “Not substantiated” (moderate); the analysis and numbers are unchanged. Appendix A lists the pledge labels.